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The Kinetics of the Redox Reactions of Ubiquinone Related to the Electron-Transport Activity in the Respiratory Chain

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The role of ubiquinone (Q) in the respiratory chain is quantitatively analyzed by correlating both the steady-state redox level of Q and the kinetics of oxidation and reduction with the flux of the overall electron transport in uncoupled submitochondrial particles. This is achieved by experimentally defining donor and acceptor activities for Q.

1. The degree of reduction of Q in the steady state is proportional to the respiratory activity with NADH and succinate, if the respiratory activity is varied by titrating the donor side for Q with rotenone and malonate, respectively. The proportionality constant (acceptor activity, V_{ox}) is independent of the substrate used.

2. The degree of oxidation of Q in the steady state is proportional to the respiratory activity as varied by titration of the acceptor side for Q with antimycin. The proportionality constant (donor activity, V_{red}) is independent of the acceptor activity and depends on the dehydrogenase activity for NADH and succinate.

3. From these experimental relations, the redox state of Q in the steady state and the respiratory activity can be described as functions of the donor and the acceptor activity only. The equations are valid with both NADH and succinate as the substrates.

4. The kinetics of oxidation of Q on the addition of oxygen as measured by the quench-flow method are in agreement with that measured by direct absorption recording in a mixing chamber. The reaction is first order with a rate constant equal to the acceptor activity divided by the amount of redox-active Q (V_{ox}/Q_a). The final steady-state level (in the present case 15% reduction) is a result also of the reduction reaction with the first-order rate constant, V_{red}/Q_a.

5. The acceptor activity can also be measured as maximum respiratory activity with durohydroquinone. This activity is independent of the presence of Q but sensitive to antimycin. Thus, the acceptor activity for ubiquinone can be measured by three independent methods.

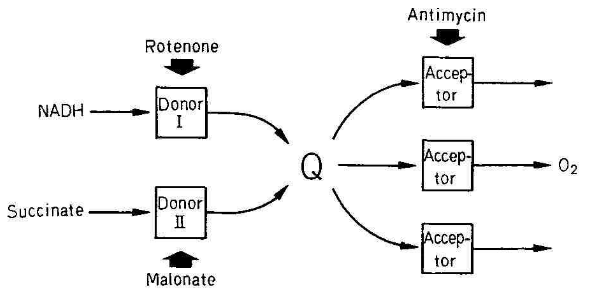
6. It is concluded, that also in the steady state the reduction and oxidation of the active Q-pool follow pseudo-first-order reactions, the rates of which are equal to the respiratory rate. The total amount of redox-active Q is kinetically and functionally homogeneous and is not divided into substrate-specific compartments.

The relation between the kinetics of the redox reactions of the mitochondrial Q and its function in electron transport has been considered so far only in qualitative terms and contradicting conclusions were drawn from these studies. Chance and Redfearn [1], Storey and Chance [2,3] and Chance *et al.* [4] concluded that Q is either on a bypass or on a blind alley of the respiratory chain whereas Klingenberg and Kröger found the redox response of Q to be consistent with that of an obligatory respiratory component [5—8]. The latter view was extended by convincing extraction-reactivation experiments by Ernster *et al.* [9].

In this communication the kinetic data of the redox reactions of Q are quantitatively related to the redox state of Q in the steady state and to the respiratory activity. This enables one to judge in a quantitative way the functional significance of the kinetics. This analysis also concerns the question whether the total content of Q is functionally homogeneous [5—8] or whether it is divided into specific compartments [10,11].

Abbreviations. Q, ubiquinone; Q_{ox}, oxidized ubiquinone; Q_{red}, reduced ubiquinone; DQH_2, durohydroquinone.

Enzymes. Catalase (EC 1.11.1.6); DT-diaphorase or reduced-NAD(P) : (acceptor) oxidoreductase (EC 1.6.99.2).



Scheme 1

In Scheme 1, Q is considered to be an obligatory respiratory component and the respiratory chain is treated as if it consists of segments which reduce (donors) and oxidize Q (acceptor). It is to be expected that the activity of the donor and of the acceptor is reflected in the redox state of Q in the steady state. The activities of the donors and of the acceptor can be varied independently by specific inhibitors and thus a variety of steady states can be adjusted in which the respiratory activity and the redox state of Q can be measured.

MATERIALS AND METHODS

Preparation of Submitochondrial Particles

Submitochondrial particles were prepared by sonication of beef heart mitochondria. 5 ml of the mitochondrial suspension containing 20 mg/ml protein were sonicated six times for 5 s at 0—5 °C with the Branson sonifier, model J17V (Danbury, Connecticut, U.S.A.), at maximal energy. Normal submitochondrial particles were obtained by sonication in 0.3 M sucrose, submitochondrial particles supplemented with cytochrome *c* with further addition of 0.3 mg/ml cytochrome *c* and submitochondrial particles deficient of cytochrome *c* in 0.1 M phosphate buffer pH 7.2. The homogenates were diluted two-fold and centrifuged for 15 min at $10000 \times g$. The supernatant was centrifuged for 1 h at $100000 \times g$, the sediment (submitochondrial particles) suspended in 0.3 M sucrose and stored in liquid nitrogen maximally one month before use. Beef heart mitochondria were prepared with bacterial proteinase (Novo, Mainz, Germany) according to Smith [12] and stored in liquid nitrogen.

The concentration of the submitochondrial particles was referred to as protein content as determined by the modified biuret method with KCN [13].

Determination of the Respiratory Activity

The respiration was recorded using a micro Clark-type oxygen electrode of our own design. The standard incubation medium was 50 mM Tris pH 7.4. 1 µmol carbonyl cyanide *p*-trifluoromethoxyphenylhydrazone/g protein was added throughout. Durohydroquinone was applied in saturated ethanolic solution (about 50 mM), which was kept anaerobic. The final concentration was about 1 mM durohydroquinone. When succinate was used, the suspension was kept anaerobic with 20 mM succinate and the respiration was initiated with catalase + H_2O_2. With NADH as the substrate 50 µM NADH was first added to the suspension and allowed to be oxidized before the respiration was started again with 2 mM NADH [14]. With these procedures the inhibition of the dehydrogenases was released to an extent, where constant respiratory activities are obtained.

Determination of the Redox State of Q in the Steady State

In the steady state of respiration 0.5 ml of the suspension was sampled from the O_2-electrode vessel and mixed with 2.5 ml of a mixture of 60% (v/v) of methanol and light petroleum (b.p. 40—60 °C). In these extracts oxidized Q ($Q_{ox} + Q_i$) was determined as described earlier [13]. As expressed by Equation (1), only part (Q_a) of the total Q (Q_t) is reducible even under equilibrium conditions in the presence of KCN and substrate [13,15], so that Q_t is divided up into an active part (Q_a) and an inactive part (Q_i):

$$Q_t = Q_a + Q_i. \tag{1}$$

Q_i varies between 10 and 20% of Q_t, it depends on the particle preparations and increases with the age of the submitochondrial particles. In the following equations only the redox state of Q_a is considered. According to Equation (2) Q_{ox} and Q_{red} can vary between zero and Q_a.

$$Q_a = Q_{ox} + Q_{red} \tag{2}$$

Q_t was determined in extracts of the submitochondrial particles without any addition and Q_i in the presence of NADH or succinate plus 1 mM KCN.

Measurement of the Kinetics of Oxidation of Q_{red}

The Moving-Mixing Chamber. The moving-mixing chamber [16] was used with the dual-wavelength spectrophotometer at 280—289 nm for measuring the oxidation of Q_{red} [13,17]. The reaction was started by the addition of standard incubation medium saturated with oxygen, to the anaerobic suspension of submitochondrial particles. The typical oscilloscope read-out given in Fig. 1 shows a dead-time of about 40 ms after the addition of the medium.

The Quench-Flow Apparatus. A quench-flow apparatus was developed for the rapid extraction

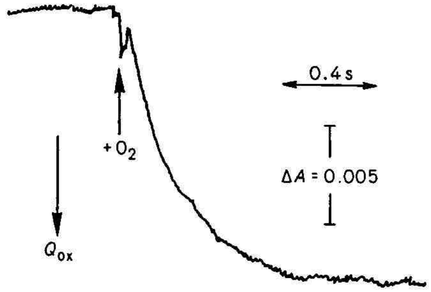


Fig. 1. *Recording of the absorption at 280–289 nm on the addition of oxygen to anaerobic submitochondrial particles with the moving-mixing chamber.* The cuvette (optical path length 0.2 cm) contained 1.5 ml of normal submitochondrial particles (2.5 mg/ml protein) in standard incubation medium with 20 mM succinate, 1 mM malonate and 1 μmol carbonyl cyanide *p*-trifluoromethoxylphenyl hydrazone per g protein under anaerobic conditions at 15 °C. The reaction was started by the addition of 15 μl of standard incubation medium saturated with oxygen at 0 °C

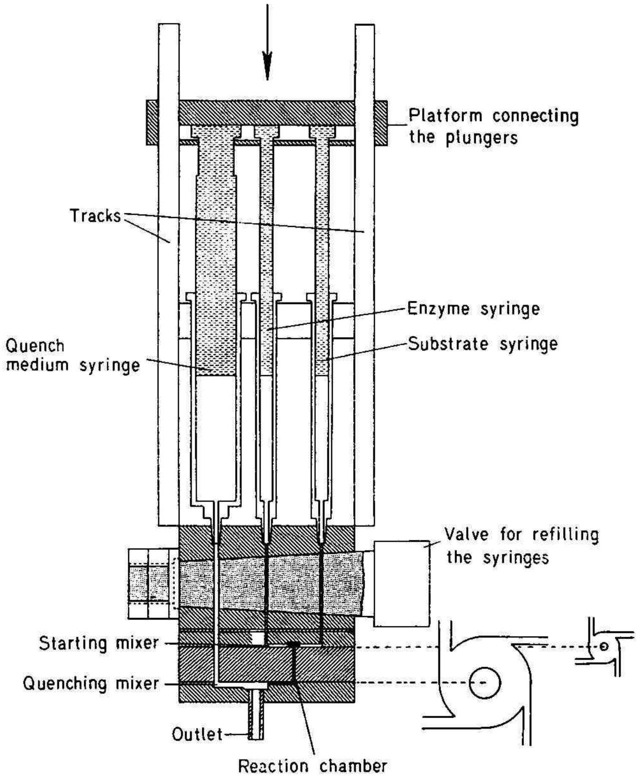


Fig. 2. *Quench-flow apparatus used for measuring the kinetics of oxidation of Q_{red}.* The apparatus consists of two 1-ml syringes (enzyme and substrate syringe) and one 10-ml quench medium syringe, the plungers of which are connected to the movable platform. The platform was pushed downward by a wheel having a cam which is machined so that the distance from the center is proportional to the angle (= one turn of an Archimedes spiral). The wheel was mounted to the axle of a gear, driven by a motor. t_P [Equation (3)] was calculated from the frequence of the wheel. The steel cones of the syringes are fitted to the polyvinylchloride block, which contains the valve made of teflon with a steel core. The starting and stopping mixers are engraved in polyvinylchloride plates. These plates together with the polyvinylchloride block carrying the reaction tube and teflon washers are fixed to the valve block by twelve screws. The interchangeable reaction tubes used were of 16-mm length and 1-mm diameter (small reaction tube) and 55-mm length and 1.8-mm in diameter (large reaction tube). An upper view of the mixers is shown in the inserts in threefold magnification. The cross section of the four tangential jets of the starting mixer is 0.12 mm², the diameter of the chamber is 3 mm and the height is 0.35 mm. The quenching chamber is 10 mm in diameter and the height is 0.5 mm. The reaction mixture jets are 0.24 mm² in cross section and the quenching medium jets 1.2 mm²

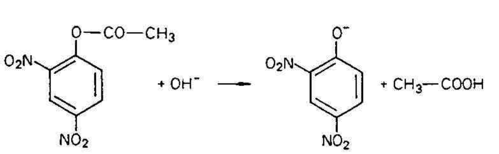
of Q and the determination of its redox state in the extract. This enables one to measure the kinetics of the oxidation of Q_{red} on the addition of oxygen. The design of the apparatus as modified from that reported in the literature [18], is given in Fig. 2. The apparatus consists of two 1-ml and one 10-ml syringes, which were filled from reservoirs (not drawn) by raising the platform connecting the plungers of the syringes with the valve in filling position. One 1-ml syringe contained the submitochondrial particles (1 mg/ml protein) in standard incubation medium made anaerobic with succinate and the other the aerated standard incubation medium. The 10-ml syringe contained the extraction medium [13] consisting of 70% (v/v) methanol and light petroleum (b.p. 40–60 °C). With the valve in operating position the platform was pushed down at a uniform rate. This causes the anaerobic suspension of the submitochondrial particles to be mixed with the aerated buffer in the starting mixer and the resulting reaction mixture to be mixed with the extraction medium in the quenching mixer. In the extracts Q_{ox} and Q_{red} were determined as described above.

The time interval between starting and stopping the reaction (t_R, reaction time) is related to the time interval used for discharging the syringes (t_P) by Equation (3):

$$t_R = v_R \cdot t_P / v_S \tag{3}$$

v_S represents the volume of the reactants in the 1-ml syringes and v_R the dead volume of the apparatus (reaction chamber) and includes the contents of the starting chamber, of the reaction tube and of the channels leading to the quenching mixer. v_R is calculated from the dimensions of the apparatus as 19.7 μl with the smaller and as 147 μl with the larger reaction tube.

The function of the quench-flow apparatus was tested using the alkaline hydrolysis of 2,4-dinitrophenolacetate [19]. This reaction (a) is pseudo-first-order with respect to the concentration of 2,4-dinitrophenolacetate, if OH^- is applied in a large excess.



(a)

In the experiment of Fig. 3, the kinetics of this reaction are followed with the quench-flow apparatus. The reaction is started by mixing 2,4-dinitrophenolacetate with OH^- coming from the enzyme syringe and the substrate syringe, respectively, and is stopped by 1.5 M HCl as the quench medium. The reaction product, dinitrophenolate, is determined photometrically, after the mixture is brought to pH 4.8 by the addition of acetate.

In Fig. 3A, the reaction parameter on the ordinate is plotted against various t_P using the small reaction tube. A linear relation passing through the origin is obtained up to $t_P = 8$ s. This corresponds to a minimal flow velocity of 1 m/s, down to which the mixers operate satisfactorily. After calculating t_R from t_P according to Equation (3) with $\nu_R/\nu_S = 0.0082$ ($\nu_S = 2.4$ ml), the pseudo-first-order rate constant of the reaction is calculated as $11.5\ s^{-1}$ from the slope of the line. With the concentration of OH^- (247.5 mM) in the reaction tube, the second-order rate constant is calculated to be $46\ M^{-1}\ s^{-1}$, which is consistent with the value reported in the literature [19]. In the experiment of Fig. 3B the large reaction tube ($\nu_R/\nu_S = 0.061$) and a lower concentration of OH^- (47.5 mM) is used. Again a linear relation is obtained up to $t_P = 8$ s and from the slope the pseudo-first-order constant is calculated to be $2.2\ s^{-1}$. On division by the concentration of OH^-, this gives the same second-order constant as obtained from the experiment of Fig. 3A. These experiments confirm the two values calculated for ν_R and ensure that reaction times between 5 ms and 0.5 s can be used.

RESULTS

Titration of the Donor Activity

In the experiment of Fig. 4 the respiratory activities with NADH and succinate of four preparations of submitochondrial particles, which differ in the acceptor activity, are "titrated" with rotenone and malonate, respectively. The respiratory activities (v_D) are plotted against the degrees of reduction of Q_a (Q_{red}/Q_a) in the corresponding steady states. The values measured for succinate are given as open and those for NADH as closed marks. The maximal respiration and the corresponding degree of reduction of Q_a with NADH and succinate are measured for each preparation in the absence of the inhibitors. As expected from Scheme 1, Q_{red} is oxidized on decreasing the donor activity. The parameters obtained for one preparation follow a linear relation, which is expressed by Equation (3a), where V_{ox} means the slope of the straight line.

$$v_D = V_{ox} \cdot Q_{red}/Q_a. \tag{3a}$$

V_{ox} represents the acceptor activity of each preparation. Accordingly V_{ox} increases with the content of cytochrome *c* of the preparations. Preparation A is obtained by sonication of the mitochondria

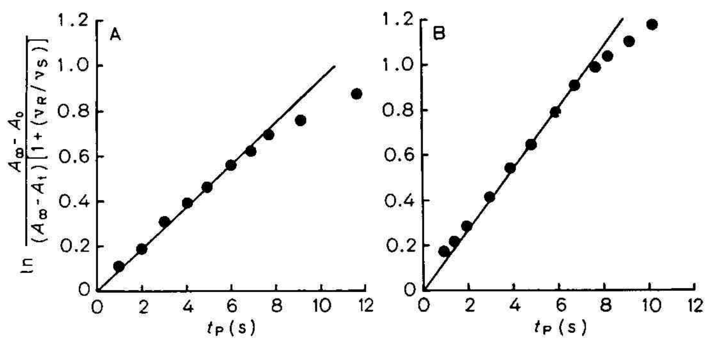


Fig. 3. *Calibration of the quench-flow apparatus.* The kinetics of the hydrolysis of 2,4-dinitrophenylacetate at 25 °C were followed using the quench-flow apparatus. The enzyme syringe was filled with 1 mM 2,4-dinitrophenylacetate in 5 mM HCl and the substrate syringe with 0.5 M KOH using the small reaction tube (A) and with 0.1 M KOH using the large reaction tube (B). The syringe for the quench medium contained 1.5 M HCl. The syringes were discharged within the various time intervals (t_P). The quenched reaction mixture was brought to pH 4.8 by the addition of about 5 ml of a solution of potassium acetate (1 g plus 1 ml water) and the absorbance at 366 nm (A_t) was measured. A_0 was obtained from an equivalent mixture to which KOH was added after the quenching medium. A_∞ was measured in a mixture, to which the quenching medium was added after complete hydrolysis of 2,4-dinitrophenylacetate. The term ν_R/ν_S corrects for the dead volume (ν_R) of the apparatus. The contents of the reaction chambers (ν_R) were calculated from the dimensions of the reaction tubes, of the starting chamber and of the channels leading from the reaction tube to the quenching chamber. ν_R was 19.7 µl with the small and 147 µl with the big reaction tube. The volume of the reaction mixture (ν_S) was 2.4 ml

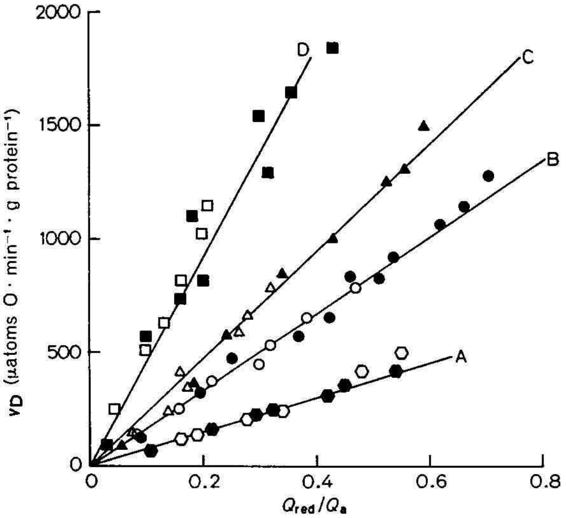


Fig. 4. *Relation of the respiratory activity to the degree of reduction of Q_a on variation of the donor activity.* The donor activity with NADH and succinate was varied by rotenone and malonate, respectively. The respiratory activity and the degree of reduction of Q_a of submitochondrial particles in standard incubation medium at 25 °C were measured as described under Methods. The maximal values measured with NADH and succinate for each preparation were obtained in the absence of the inhibitors. Experimental conditions: (A) cytochrome-*c*-deficient submitochondrial particles (1.05 mg/ml) with NADH (●) and succinate (○) as the substrate, $Q_a = 3.75$ μmol/g protein; (B) normal submitochondrial particles, 0.5 mg/ml with NADH (●) and 0.7 mg/ml with succinate (○) as substrate, $Q_a = 4.7$ μmol/g protein; (C) normal submitochondrial particles, 0.5 mg/ml with NADH (▲) and 0.7 mg/ml with succinate (△), $Q_a = 3.6$ μmol/g protein; (D) cytochrome-*c*-supplemented submitochondrial particles (0.5 mg/ml) with NADH (■) and succinate (□), $Q_a = 3.5$ μmol/g protein

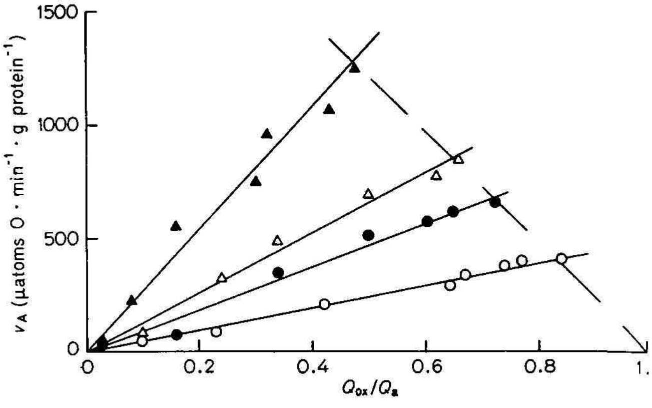


Fig. 5. *Relation of the respiratory activity to the degree of oxidation of Q_a on variation of the acceptor activity.* The acceptor activity was varied by antimycin. The points on the dashed line were obtained in the absence of antimycin. Normal submitochondrial particles (Preparation C of Fig. 4) were incubated in standard incubation medium at 25 °C. The respiratory activity and the degree of oxidation of Q_a were determined as described under Methods. Experimental conditions: (▲) 0.5 mg/ml submitochondrial particles and NADH as the substrate; (△) 1 mg/ml submitochondrial particles and NADH plus 0.084 μmol/g protein rotenone; (●) 1 mg/ml submitochondrial particles and succinate; (○) 1 mg/ml submitochondrial particles and 20 mM succinate plus 0.3 mM malonate

in phosphate buffer, so that the submitochondrial particles are depleted from cytochrome *c*, and a minimum value for V_{ox} (760 μatoms O × min^{-1} × g $protein^{-1}$) is obtained. Sonication of the mitochondria in sucrose solution gives submitochondrial particles, which retain most of the cytochrome *c* (preparations B and C). The V_{ox} of these preparations is 1700 and 2400 μatoms O × min^{-1} × g $protein^{-1}$, respectively. The maximum value for V_{ox} (4600 μatoms O × min^{-1} × g $protein^{-1}$) is found with a preparation sonicated in the presence of sucrose and cytochrome *c* (preparation D).

The parameters measured with NADH and succinate for the same preparation fit the same straight line. This confirms, that the acceptor activities are given by the slopes of the lines, since the acceptor activity should be independent of the donor according to scheme 1. The lower donor activity with succinate than with NADH is reflected in the lower respiratory activity in the absence of inhibitor and in the correspondingly lower degree of reduction of Q_a with succinate.

Titration of the Acceptor Activity

Preparation C of Fig. 4 is subjected to titration with antimycin in the experiment of Fig. 5. The respiratory activity (v_A) is plotted against the degree of oxidation of Q_a with four different donor activities. These are adjusted by partial inhibition of the respiration with NADH and succinate by rotenone and malonate, respectively. The rates without antimycin fit a line (dashed) originating at $Q_{ox}/Q_a = 1$. This line is identical with line C of Fig. 4. As expected from Scheme 1, Q_{ox} is reduced on the inhibition of the acceptor by antimycin and again a linear relation of the respiratory activity to the redox state of Q_a is obtained. This relation is expressed by Equation (4), in which V_{red} designates the slope of the line.

$$v_A = V_{red} \cdot Q_{ox}/Q_a \quad (4)$$

According to the four donor activities used, four different slopes are obtained (μatoms O × min^{-1} × g $protein^{-1}$): $V_{red} = 2720$ for NADH, $V_{red} = 950$ with succinate, $V_{red} = 1320$ for NADH plus rotenone and $V_{red} = 490$ for succinate plus malonate. At the four points, where the antimycin titration curves cut the dashed line, v_D equals v_A and both V_{red} given by the titration with antimycin (solid lines) and V_{ox} determined by the titration curve of the donor activity (dashed line) are valid. From Equations (3a) and (4) follows:

$$Q_{red}/Q_{ox} = V_{red}/V_{ox} \quad (5)$$

Q_{red} in Equation (5) can be substituted for by Equation (2):

$$Q_{ox}/Q_a = V_{ox}/(V_{red} + V_{ox}) \quad (6)$$

The combination of Equations (4) and (6) gives:

$$v_D = v_A = V_{red} \cdot V_{ox}/(V_{red} + V_{ox}) \quad (7)$$

In these equations the redox state of Q_a in the steady state and the respiratory activity are given as functions of the donor and acceptor activity only. These equations are demonstrated above to be valid with NADH and succinate as the substrates with and without the corresponding inhibitors of the donor activities. The equations are also valid in the presence of antimycin (unpublished results).

The Oxidation of Q_{red}

In Equations (3a) and (4) the respiratory activity is related by first-order rate equations to Q_{red} and Q_{ox} respectively, and V_{ox}/Q_a and V_{red}/Q_a can be considered as first-order rate constants. Therefore, the oxidation and reduction of Q_a can be expected to follow first-order reactions. This is confirmed by the experiment of Fig. 6. The respiratory activity with succinate of submitochondrial particles is adjusted to 90 µatoms O × min^{-1} × g protein^{-1} by the addition of malonate. Subsequently the oxidation of Q_{red} caused by the addition of oxygen to the submitochondrial particles in the anaerobic state, is measured in parallel experiments by two different methods: (a) by recording of the absorption increase at 280—289 nm caused by the rapid addition of oxygen in the moving-mixing chamber (triangles) and (b) by measuring Q_{ox} in extracts which are obtained with the quench-flow apparatus after various reaction times with O_2 (dots). The same kinetics are obtained by both methods, thus confirming that by the spectrophotometric recording the oxidation of Q_{red} is measured.

The kinetics found experimentally are compared with two theoretical curves which are drawn according to Equation (8b). This equation results from Equation (8a) on integration.

$$dQ_{ox}/dt = (V_{ox} \cdot Q_{red}/Q_a) - (V_{red} \cdot Q_{ox}/Q_a) \quad (8a)$$

$$\ln \frac{V_{ox}}{V_{ox} - [(V_{red} + V_{ox})\, Q_{ox}/Q_a]} = \left(\frac{V_{red} + V_{ox}}{Q_a}\right) \cdot t. \quad (8b)$$

The equations presume, that the oxidation of Q_{red} on the transition from the anaerobic to the steady state is a first order reaction with the same V_{ox} as obtained from the relation of the respiratory activity to the redox state of Q_a [Equation (3a)]. Simultaneous with the oxidation reaction the reduction of Q_{ox} by succinate is assumed to occur [negative term in Equation (8a)]. This reaction should also proceed according to the first order and with the value for V_{red} equal to that of Equation (4). Therefore, under the condition of the steady state when $dQ_{ox}/dt = 0$, Equation (8a) reduces to Equation (5).

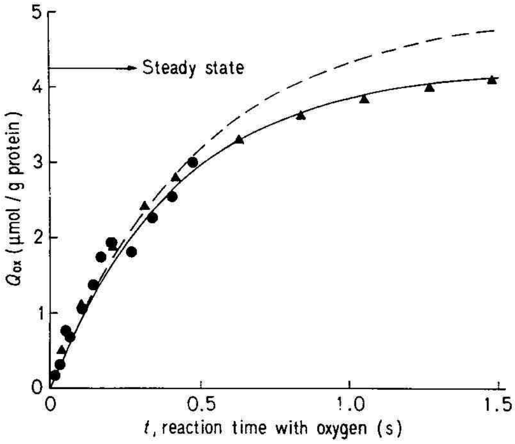


Fig. 6. *The kinetics of the oxidation of Q_{red} on the transition from the anaerobic to the steady state.* Normal submitochondrial particles were kept anaerobic in standard incubation medium with 20 mM succinate, 1 mM malonate and 1 µmol/g protein carbonyl cyanide *p*-trifluoromethoxyphenylhydrazone at 15 °C. (▲) Values were obtained from the absorption increase recorded at 280—289 nm after the addition of aerated standard incubation medium with the moving-mixing chamber (see Methods). (●) Values were obtained with the quench-flow apparatus, which gives Q-containing extracts after aeration for the various time intervals (see Methods). The respiratory activity at 15 °C with 20 mM succinate + 1 mM malonate was 90 and with DQH_2 600 µatoms O × min^{-1} × g protein^{-1}. $Q_t = 6.6$, $Q_a = 5.0$ µmol/g protein. The solid curve was drawn according to Equation (8b) with $V_{ox} = 600$ and $V_{red} = 105$ and the dashed curve with $V_{ox} = 600$ and $V_{red} = 0$ µatoms O × min^{-1} × g protein^{-1}

The solid curve in Fig. 6 is calculated according to Equation (8b) with $V_{ox} = 600$, $V_{red} = 105$ µatoms O × min^{-1} × g protein^{-1} and $Q_a = 5.0$ µmol/g protein. The value for V_{ox} of this preparation of submitochondrial particles is measured by titration of the donor activity as described in Fig. 4. V_{red} is calculated from the respiratory rate with succinate plus malonate (90 µatoms O × min^{-1} × g protein^{-1}) and V_{ox} according to Equation (7). Q_a is determined by extraction. The agreement of the actual kinetics with the theoretical curve shows that the oxidation reaction of Q_{red} on the transition from the anaerobic to the steady state is well described by Equation (8b). This does not only confirm the first-order nature of the reaction but also the value for V_{ox}, obtained from the steady-state experiment (*cf.* Fig. 4) according to Equation (3a). V_{ox} is equal to the initial rate of oxidation of Q_{red} as follows from Equation (8a) for Q_{ox} approaching zero and $Q_{red} = Q_a$:

$$(dQ_{ox}/dt)_{t \to 0} = V_{ox}: \quad (9)$$

Table 1. *Respiratory activity with DQH_2 on depletion and reincorporation of Q*

Lyophilized, Q-depleted and Q-reincorporated submitochondrial particles were prepared from normal submitochondrial particles according to Ernster *et al.* [9]. The respiratory activity with NADH and the content of Q (Q_t) were measured as described under Methods. The respiratory activity with DQH_2 was measured in a regenerating system using NADH and DT-diaphorase in the presence of rotenone. The various preparations of submitochondrial particles were suspended in standard incubation medium at 25 °C with 2 µmol rotenone/g protein, 0.1 mM duroquinone, 1 mg/ml DT-diaphorase [24] and 2 mM NADH. Q-9, ubiquinone with nine isoprene units

Preparation of submitochondrial particles	Respiratory activity with		Antimycin titer	Q_t
	DQH_2	NADH		
	µatoms O × min⁻¹ × g protein⁻¹		µmol/g protein	
Normal	810	1800	0.37	—
Lyophilized	770	1350	0.38	5.8
Q-depleted	630	0	0.36	0.04
Q-depleted + Q-9	1030	750	0.40	17.5

The first-order nature of the reduction reaction as given by Equation (4) is confirmed here by the good fit of the solid line with the data. The reaction does not fit the dashed line corresponding to Equation (8b) under the assumption that $V_{red} = 0$. The deviation is due to the reduction reaction, which is accounted for by the negative term in Equation (8a).

The Respiration with DQH_2

Externally added hydroquinones like DQH_2 are rapidly oxidized by mitochondria and submitochondrial particles [21—23]. DQH_2 reacts presumably with the same acceptor as Q_{red}, since the respiratory activity with DQH_2 is as sensitive to antimycin as that with NADH and succinate and is independent of the presence of Q.

In the experiment of Table 1, submitochondrial particles were subjected to the extraction procedure for Q of Ernster *et al.* [9].

Lyophilisation of the particles before the extraction causes a decrease of the respiratory activity with NADH. This is probably due to partial damage of the preparation. The subsequent extraction of 97% of the total Q results in the complete inhibition of the respiratory activity with NADH and the reincorporation of Q reactivates the activity. In contrast, the respiratory activity with DQH_2 is almost unaffected by the extraction of Q. Also the amount of antimycin required for maximum inhibition (antimycin-titer) is not changed by the extraction-reincorporation procedure.

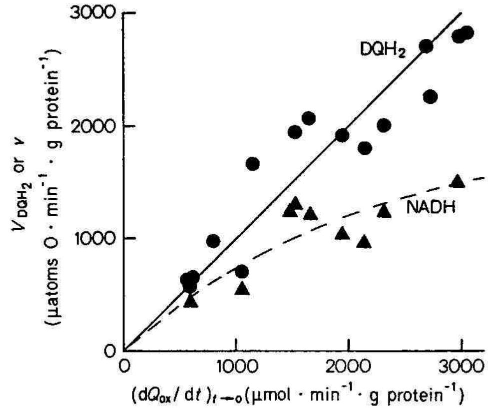


Fig. 7. *Comparison of the initial rate of oxidation of Q_{red} to the respiratory activity with DQH_2 (●) and NADH (▲).* The initial rates of oxidation of Q_{red} on the transition from the anaerobic to the steady state, $(dQ_{ox}/dt)_{t\to0}$ of various preparations of submitochondrial particles were measured with the quench-flow apparatus as in the experiment of Fig. 6. The respiratory activities with DQH_2 (V_{DQH_2}) and NADH (v) were determined as described under Methods. The solid line was drawn for better comparison and indicates equal values for $(dQ_{ox}/dt)_{t\to0}$ and V_{DQH_2}. The dashed curve gives the respiratory activity as the function of the acceptor activity according to Equation (7) with $V_{red} = 3000$ µatoms O × min⁻¹ × g protein⁻¹

The respiratory activity of submitochondrial particles with DQH_2 reaches saturation (V_{DQH_2}) at high concentration of DQH_2. In Fig. 7, V_{DQH_2} of various preparations of submitochondrial particles, with different V_{ox} due to varying contents of cytochrome *c*, are plotted against the initial rates of oxidation of Q_{red} measured by the quench-flow technique. The straight line is drawn for equal values of both parameters. The average deviation of the experimental data from the line is ± 14%, a good fit in view of the disparity of both parameters. This demonstrates that V_{DQH_2} is nearly equal to the initial rate of oxidation of Q_{red}.

For some of the preparations also the respiratory activity with NADH (v) is measured and plotted against the initial rate of oxidation of Q_{red} in Fig. 7. v is always smaller than V_{ox} and the difference between v and V_{ox} increases with V_{ox}. This agrees with Equation (7), according to which the curve is calculated with $V_{red} = 3000$ µatoms O × min⁻¹ × g protein⁻¹ as the average value of V_{red} with NADH in the various preparations (*cf.* Fig. 5). The relatively large deviation of some of the points from the dashed curve is to be expected because of the scattering of V_{red} of the different preparations around the average value. The consistency in principle of the calculated curve with the experimental data confirms the relation between v and V_{ox} given by Equation (7). Further confirmation of Equation (7) has been obtained (unpublished results) by varying V_{ox} with antimycin.

DISCUSSION

Q as an Obligatory Respiratory Component

The oxidation reaction of Q_{red} on aeration of anaerobic submitochondrial particles (Fig. 6) is first order with a reaction constant equal to V_{ox}/Q_a [Equation (8a) with $V_{red} = 0$]. The same constant is found in the steady state for the relation of the respiratory activity to Q_{red}, if the donor activity is varied [Equation (3a), Fig. 4]. This shows that the oxidation rate of Q_{red} in the steady state (and consequently the reduction rate of Q_{ox}) is equal to the respiratory rate [Equation (10)].

$$(\mathrm{d}Q_{ox}/\mathrm{d}t)_{st.\,st.} = v = (\mathrm{d}Q_{red}/\mathrm{d}t)_{st.\,st.} \tag{10}$$

According to this relation Q_a fulfils the kinetic requirements of an obligatory respiratory component. The possibility is excluded that only part of the reducing equivalents of the substrates are transferred to oxygen *via* Q [23, 25–27].

There are three independent methods to determine the acceptor activity (V_{ox}) for Q. Firstly, it is obtained from the relation between respiratory activity (as varied by the donor activity) and the degree of reduction of Q_a [Equation (3a), Fig. 4]. Secondly, V_{ox} is measured as the initial rate of oxidation of Q_{red} on transition from the anaerobic to the steady state [Equation (9), Fig. 6]. By a third method V_{ox} is measured as the rate of oxygen uptake with DQH_2 as donor (Fig. 7).

There is only one method to determine the donor activity (V_{red}) for Q: The relation between respiratory activity as varied by the acceptor activity, and the degree of oxidation of Q_a [Equation (4), Fig. 5]. Moreover there is a relation between V_{red}, V_{ox} and the respiratory activity in a given steady state [Equation (7)] which can be used as a fourth or second method for determining V_{ox} or V_{red} respectively.

The direct measurement of the reduction of Q_{ox} in order to determine the rate constant V_{red}/Q_a has not yet been possible. The initial rate of reduction, which is observed on the addition of substrate in the presence of a terminal inhibitor, is distinctly slower than V_{red}. There is probably a lag period required for the activation of the dehydrogenases by the substrates [14]. Klingenberg and Kröger [5] found initial rates of reduction of Q_{ox} equal to the respiratory rates and Chance and Redfearn [1], Chance [25], Storey and Chance [2] reported initial rates, which are even slower than the overall rates.

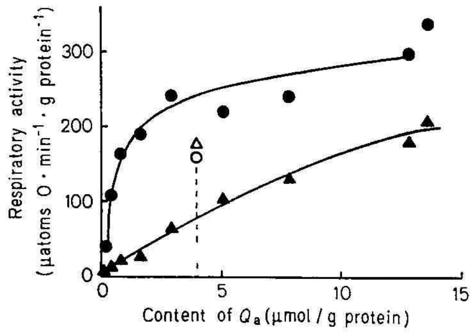


Fig. 8. *The relation between respiratory activity with succinate (●) and NADH (▲) and the content of Q_a.* Q-depleted and Q-reincorporated particles were prepared from normal submitochondrial particles according to Ernster *et al.* [9]. The amount of Q incorporated was varied by shaking about 20 mg of the depleted particles with 0.5 ml of a pentane-solution of Q-9 (ubiquinone with 9 isoprene units), the concentration of which ranged between 0.1 and 10 mM. The respiratory activity of the various preparations and Q_a were determined as described under Methods. The open marks refer to the lyophilized preparation (controls)

The possibility that Q is on a blind alley of the chain in a rapid equilibrium with an electron carrier, is excluded by the results of the extraction-reincorporation technique. With this method it was demonstrated that the presence of Q is required for both the respiration with NADH and succinate [9]. This result was contradicted recently by Albracht and Slater [28], who reported that a mitochondrial compound, which is thought to be different from Q, reactivates specifically the respiration with succinate in particles depleted from Q. However, the identity of the compound is unknown and the amount of Q present in the depleted particles was not tested in these experiments.

The experiment of Fig. 8 suggests that the reactivation of the respiration with succinate is caused by relatively small amounts of Q and not by another compound. In this experiment increasing amounts of Q are incorporated into Q-depleted particles and the portion of incorporated Q which is redox-active (Q_a) is plotted against the respiratory activites with NADH and succinate. The "depleted" particles (the first points in Fig. 8), contain about 3% of the original content of Q. This amount cannot be extracted by pentane but only by the methanol—light petroleum mixture used for quenching and for the extraction of both reduced and oxidized Q. In addition, the depleted particles retain a small respiratory activity with succinate but not with NADH. The respiratory activity with succinate is restored to the original level already by the incorporation of 15% of the original content of Q. These observations suggest that the respiratory activity with succinate is also dependent on the presence of Q as is that with

NADH. The complete reactivation of the respiration with NADH requires about three times the original content of redox-active Q. This difference in the restoration of the two activities was overlooked by Albracht and Slater [28] and also by Gutman and Singer [11] because these authors refer the respiratory activity to the total amount of Q incorporated and not to the active part (Q_a). The requirement of an excess of Q for the reactivation of the respiration with NADH is probably due to an increase of the K_m of the NADH dehydrogenase for Q as an artifact of the pentane-extraction (see below).

The First-Order Nature of the Redox Reactions of Ubiquinone

Although the electron transfer in the respiratory chain is a bimolecular process, the transitions of the respiratory carriers from the reduced to the oxidized state closely follow first-order reactions and have been treated as such [4,5]. This is feasible for a chain of fixed carriers from which the electrons are drained towards oxygen. It is, however, difficult to rationalize that a pool of movable hydrogen-carrier molecules such as Q also follows a first-order reaction. Instead, a mixed zero- and first-order reaction is expected since Q can be assumed to saturate its donating and accepting component (Fig. 8).

As a plausible explanation, the oxidation reaction of Q_{red} is assumed to reflect the oxidation of a subsequent fixed carrier (component B in Scheme 2).

$$SH_2 \dashrightarrow D \rightarrow Q \rightleftharpoons B \rightarrow C \dashrightarrow O_2.$$

This would hold if B and Q are in a close equilibrium, which is adjusted faster than B_{red} is oxidized. The larger pool size of Q in equilibrium with B as compared to that of C would cause C to reach its steady-state level almost before the oxidation of B_{red} starts. As a result, the oxidation of B_{red} (and consequently that of Q_{red}) would follow a pseudo-first-order reaction. The assumption that the oxidation of Q_{red} by B_{ox} is not rate-limiting, is in agreement with the observation, that the respiratory activity with DQH_2 at saturation is equal to V_{ox}.

The properties postulated for component B and C are consistent with those of the *b*- and *c*-cytochromes, respectively. The redox potentials of Q and of the *b*-cytochromes are reported to be almost equal under uncoupled conditions [15] and this would enable a close equilibrium between these components. The half time of oxidation of the *b*-cytochromes [4,29], which is in the order of magnitude (about 0.1 s) of that of Q, may indicate that the equilibrium is adjusted faster than the *b*-cytochromes are oxidized. The half-time of oxidation of the *c*-cytochromes (about 5 ms) is considerably faster than those of Q and the *b*-cytochromes [4,29].

It appears necessary here to repeat [5,7,8] for the benefit of the less initiated reader that one reason for the delay in accepting the role of Q in the respiratory chain has been errors in the evaluation of the kinetics of the respiratory chain. Thus, the sequence of the components was ordered according to the sequence of the half-times of oxidation after the additional of an oxygen pulse, this being done disregarding the size of the electron reservoirs [3,4], instead of evaluating the actual reaction rates. This led to the exclusion of Q from the main electron path. Correspondingly the comparatively long half-time of oxidation of the *b*-cytochromes has been misunderstood [4], because the extended pool-size caused by the equilibrium with Q was disregarded.

The Forms of Q Involved in the Redox Reactions

The rate equation for the oxidation and reduction of Q (Equation 9a) indicates that Q_{red} and Q_{ox} donate and accept the reducing equivalents in the respiratory process. In contrast, the redox reactions of plastoquinone in the photosynthetic process are reported to involve radicals as the donor and the acceptor [39]. This was deduced from a slightly sigmoidal shape of the kinetics of the redox reactions of plastoquinone and is unconvincing. From the first-order nature of the redox reactions of Q_a it can be excluded, that the radicals of Q react with the donor and the acceptor of Q. However, it is possible that the radicals are formed as intermediary products of the reduction of Q_{ox} (Reaction b) and of the oxidation of Q_{red} (Reaction c).

$$D_{red} + Q_{ox} + H^+ \rightarrow {}^{\cdot}QH + D_{ox} \tag{b}$$

$$Q_{red} + B_{ox} \rightarrow {}^{\cdot}QH + B_{red} + H^+. \tag{c}$$

The transfer of single electrons is probable, especially since iron-sulphur proteins and a cytochrome are likely candidates for the components D and B of Scheme 2, which directly reduce and oxidize Q. ${}^{\cdot}QH$ is probably the more stable radical, since the dissociation constant of the H^+ is 0.5 μM [31] and its basic properties appear to be very weak [32]. As demonstrated by electron-spin-resonance spectroscopy, the amount of the radicals within the mitochondrial membrane does not exceed 1.5% of the total Q under equilibrium conditions in the presence of KCN and substrate [33]. This amount cannot be differentiated by the methods used here. The steady-state concentration is not expected to be higher, since the disproportionation of the radicals (Reaction d) is much faster than their formation by electron transport.

$$2\,{}^{\cdot}QH \rightarrow Q_{red} + Q_{ox} \tag{d}$$

The fastest reaction, according to which radicals could be formed is the oxidation of Q_{red} on the transition from the anaerobic to the steady state. The maximal pseudo-first-order rate constant of this reaction was measured to be $15 s^{-1}$ if Q_{ox} is formed (Fig. 4 and 7). This is equivalent to a value of $30 s^{-1}$ for the conversion of Q_{red} to ˙QH with a corresponding half-time of formation of the radicals of 23 ms. For comparison the rate constant of Reaction (d) was measured to be $4.8 \cdot 10^7 M^{-1} s^{-1}$ in methanol [31]. From this value the half-time of disproportionation is calculated to be 0.36 ms, assuming a radical concentration of 40 μM within the membrane, which corresponds to about 1 % of the content of Q.

The intermediary appearance of relatively high amounts of other derivatives of Q different from Q_{red} and Q_{ox} appears to be unlikely. The kinetics of the absorption increase of the particles measured at 280—289 nm on the transition from the anaerobic to the steady state (Fig. 3) is quantitatively attributed by the extraction method to the conversion of Q_{red} to Q_{ox} on the basis of the difference absorption coefficient of Q. Therefore, only derivatives of Q, which are converted to Q_{ox} on the extraction and have about the same absorption coefficient, would not be differentiated.

The Homogeneity of the Q-Pool

10—20 % (Q_i) of the total Q (Q_t) in the submitochondrial particles [Equation (1)] is inactive, *i.e.*, not reducible by the substrates [15] and is therefore disregarded in the kinetic calculations. The active part (Q_a) responds in a kinetically homogeneous fashion. This is demonstrated by the oxidation reaction of Q_{red}, which occurs on aeration of the anaerobic submitochondrial particles and by the finding, that the redox state of Q_a is a function of the donor- and acceptor activity only (Equation 6). The finding that Equation (6) is valid for the total Q_a with succinate and NADH as the substrates, demonstrates that each molecule of Q_a is accessible to the reducing equivalents of both dehydrogenases. This conclusion is in contrast to the compartmentation postulated from depletion-reactivation studies [10, 11]. The fact that the respiration with NADH exhibits a greater specificity for the homologues of Q [10] and requires greater amounts of Q for reactivation than the respiration with succinate [11], was taken as evidence for the compartmentation of Q. This interpretation is based on the assumption that each molecule of Q has a fixed position in the membrane. On the basis that Q diffuses within the membrane, however, the observations mentioned can be easily explained by enzyme-specificity and different K_m values of the dehydrogenases for Q. The diffusion-hypothesis appears to be more probable in view of the fact that one molecule of a dehydrogenase can reduce 50—100 molecules of Q [7]. Furthermore, the finding that more than three times the norma. content of Q can be reincorporated into the meml brane and is reducible by one substrate (Fig. 8)-argues against a fixed position of Q in the membrane.

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